

✓ The "Sugar Trap" Market Gap Analysis

Client: Helix CPG Partners

This notebook is a starter scaffold covering:

1. Data ingestion (subset of Open Food Facts)
2. Data cleaning
3. Category bucketing
4. Nutrient Matrix scatter plot
5. Placeholder sections for the recommendation, ingredient analysis, and your Candidate's Choice addition.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

pd.set_option('display.max_columns', 50)
```

✓ 1. Data Ingestion and Data Cleaning(Story 1)

Download the Open Food Facts CSV export. We only pull the columns we need and cap rows so this stays reproducible and fast.

- Drop rows missing `product_name`, `sugars_100g`, or `proteins_100g`
- Filter biologically impossible values (nutrient-per-100g must be between 0 and 100)

```
import pandas as pd

!wget -q https://openfoodfacts-ds.s3.eu-west-3.amazonaws.com/en.openfoodfacts.org.products.csv.gz -O off_products.csv.gz
!gunzip -f off_products.csv.gz

COLUMNS_NEEDED = [
    'product_name',
    'categories_tags',
    'ingredients_text',
    'sugars_100g',
    'proteins_100g',
    'fat_100g',
    'fiber_100g'
]

df = pd.read_csv(
    'off_products.csv',
    sep='\t',
    usecols=lambda c: c in COLUMNS_NEEDED,
    nrows=500_000,
    low_memory=False
)

clean = df.copy()

# Drop rows missing critical fields
clean = clean.dropna(subset=['product_name', 'sugars_100g', 'proteins_100g'])

# Filter impossible nutrient values (0-100g per 100g of product)
for col in ['sugars_100g', 'proteins_100g', 'fat_100g', 'fiber_100g']:
    if col in clean.columns:
        clean = clean[(clean[col].isna()) | ((clean[col] >= 0) & (clean[col] <= 100))]

print(f'Rows before cleaning: {len(df):,}')
print(f'Rows after cleaning: {len(clean):,}')
clean.head()
```

Rows before cleaning: 500,000
 Rows after cleaning: 109,624

	product_name	categories_tags	ingredients_text	fat_100g	sugars_100g	fiber_100g	proteins_100g
689	Pinto Bean	en:asian-style-ready-meal	NaN	10.20000	4.900000	3.800000	17.500000
690	Croquetas de bacalao	NaN	NaN	12.10000	1.900000	NaN	5.900000
691	Keto & GF Granola	NaN	NaN	54.83871	3.225806	9.677419	19.677419
797	Ben's Pure Maple Cream	NaN	NaN	30.28000	37.720000	NaN	5.680000
829	Guimauve chocolat smarties	NaN	NaN	19.00000	65.000000	NaN	5.700000

3. Category Wrangler (Story 2)

Parse `categories_tags` and assign each product a readable Primary Category using keyword matching.

TODO: Tune the keyword lists below against your actual data, inspect the most common tags first.

```
# Inspect the most common raw tags to inform your keyword buckets
clean['categories_tags'].dropna().str.split(',').explode().str.strip().value_counts().head(30)
```

	count
en:plant-based-foods-and-beverages	21737
en:plant-based-foods	20161
en:cereals-and-potatoes	12513
en:snacks	9061
en:breads	7152
en:dairies	6794
en:sweet-snacks	6036
en:fermented-foods	5959
en:fermented-milk-products	5905
en:desserts	5374
en:cereals-and-their-products	5048
en:dairy-desserts	4106
en:fermented-dairy-desserts	4024
en:yogurts	3969
en:beverages	3690
en:condiments	3628
en:breakfasts	3343
en:fruits-and-vegetables-based-foods	3041
en:sauces	2993
en:meals	2752
en:biscuits-and-cakes	2613
en:breakfast-cereals	2481
en:meats-and-their-products	2358
en:confectioneries	2220
en:salty-snacks	2181
en:frozen-foods	2156
en:appetizers	1991
en:spreads	1916
en:groceries	1862
en:cheeses	1797

```
CATEGORY_KEYWORDS = {
    'Cookies & Biscuits': ['cookie', 'biscuit'],
    'Chips & Savory Snacks': ['chip', 'crisp', 'pretzel', 'popcorn'],
    'Bars': ['bar', 'granola-bar', 'protein-bar'],
```

```

'Cereal & Granola': ['cereal', 'granola', 'muesli'],
'Candy & Confectionery': ['candy', 'chocolate', 'sweet', 'confectionery'],
}

def assign_category(tags):
    if pd.isna(tags):
        return 'Other'

    # Explicit parsing of the comma-separated categories_tags string into a list
    # e.g. "en:snacks,en:sweet-snacks,en:biscuits" -> ['en:snacks', 'en:sweet-snacks', 'en:biscuits']
    tag_list = [t.strip().lower() for t in tags.split(',')]

    for category, keywords in CATEGORY_KEYWORDS.items():
        if any(kw in tag for kw in keywords) for tag in tag_list:
            return category
    return 'Other'

# Ensure 'primary_category' is (re)created using the corrected parsing logic
clean['primary_category'] = clean['categories_tags'].apply(assign_category)
clean['primary_category'].value_counts()

```

	count
primary_category	
Other	87426
Cereal & Granola	11642
Candy & Confectionery	4782
Cookies & Biscuits	3121
Chips & Savory Snacks	1573
Bars	1080

dtype: int64

4. Nutrient Matrix Visualization (Story 3)

Scatter plot: Sugar (X) vs Protein (Y), colored by category. Look for the empty quadrant.

```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Data ingestion (from fqs8JXH9MrdZ and T8x2jwfXMrDg)
!wget -q https://openfoodfacts-ds.s3.eu-west-3.amazonaws.com/en.openfoodfacts.org.products.csv.gz -O off_products.csv.gz
!gunzip -f off_products.csv.gz

COLUMNS_NEEDED = [
    'product_name',
    'categories_tags',
    'ingredients_text',
    'sugars_100g',
    'proteins_100g',
    'fat_100g',
    'fiber_100g'
]

df = pd.read_csv(
    'off_products.csv',
    sep='\t',
    usecols=lambda c: c in COLUMNS_NEEDED,
    nrows=500_000,
    low_memory=False
)

# Data cleaning (from T8x2jwfXMrDg)
clean = df.copy()

# Drop rows missing critical fields
clean = clean.dropna(subset=['product_name', 'sugars_100g', 'proteins_100g'])

# Filter impossible nutrient values (0-100g per 100g of product)
for col in ['sugars_100g', 'proteins_100g', 'fat_100g', 'fiber_100g']:
    if col in clean.columns:
        clean = clean[(clean[col].isna()) | ((clean[col] >= 0) & (clean[col] <= 100))]

# Category assignment (from d8YwK1a-Mrds)
CATEGORY_KEYWORDS = {

```

```

'Cookies & Biscuits': ['cookie', 'biscuit'],
'Chips & Savory Snacks': ['chip', 'crisp', 'pretzel', 'popcorn'],
'Bars': ['bar', 'granola-bar', 'protein-bar'],
'Cereal & Granola': ['cereal', 'granola', 'muesli'],
'Candy & Confectionery': ['candy', 'chocolate', 'sweet', 'confectionery'],
}

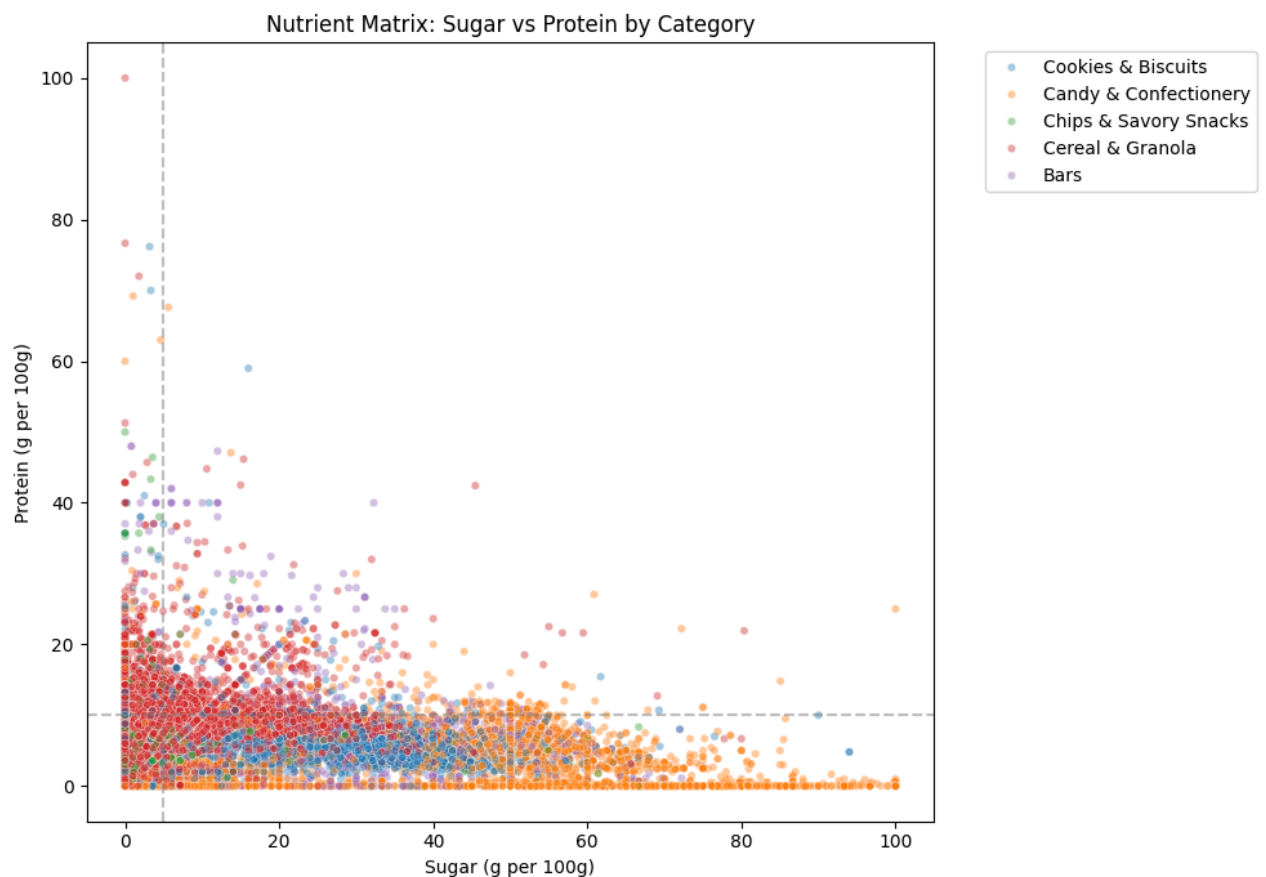
def assign_category(tags):
    if pd.isna(tags):
        return 'Other'
    tags_lower = tags.lower()
    for category, keywords in CATEGORY_KEYWORDS.items():
        if any(kw in tags_lower for kw in keywords):
            return category
    return 'Other'

# Ensure 'primary_category' is created if not already present
if 'primary_category' not in clean.columns:
    clean['primary_category'] = clean['categories_tags'].apply(assign_category)

plot_df = clean[clean['primary_category'] != 'Other']

plt.figure(figsize=(10, 7))
sns.scatterplot(
    data=plot_df,
    x='sugars_100g',
    y='proteins_100g',
    hue='primary_category',
    alpha=0.4,
    s=20
)
plt.title('Nutrient Matrix: Sugar vs Protein by Category')
plt.xlabel('Sugar (g per 100g)')
plt.ylabel('Protein (g per 100g)')
plt.axvline(5, linestyle='--', color='gray', alpha=0.5) # candidate 'low sugar' threshold
plt.axhline(10, linestyle='--', color='gray', alpha=0.5) # candidate 'high protein' threshold
plt.legend(bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()

```



5. The Recommendation (Story 4)

Use the chart above to identify the empty quadrant, then quantify it below.

```
# Example: how many products currently sit in the 'high protein, low sugar' quadrant, by category?
HIGH_PROTEIN_THRESHOLD = 10
LOW_SUGAR_THRESHOLD = 5

empty_quadrant = plot_df[
    (plot_df['proteins_100g'] >= HIGH_PROTEIN_THRESHOLD) &
    (plot_df['sugars_100g'] <= LOW_SUGAR_THRESHOLD)
]

quadrant_counts = plot_df.assign(
    in_empty_quadrant=(plot_df['proteins_100g'] >= HIGH_PROTEIN_THRESHOLD) &
    (plot_df['sugars_100g'] <= LOW_SUGAR_THRESHOLD)
).groupby('primary_category')['in_empty_quadrant'].mean().sort_values()

print('Share of products already in the high-protein/low-sugar quadrant, by category:')
quadrant_counts
```

Share of products already in the high-protein/low-sugar quadrant, by category:

primary_category	in_empty_quadrant
Candy & Confectionery	0.012756
Bars	0.055556
Chips & Savory Snacks	0.064209
Cookies & Biscuits	0.086511
Cereal & Granola	0.195070

dtype: float64

Key Insight (fill in once you've run the analysis):

"Based on the data, the biggest market opportunity is in **[Category Name]**, specifically targeting products with **[X]g** of protein and less than **[Y]g** of sugar."

6. Bonus: Hidden Gem — Top Protein Sources

Analyze `ingredients_text` for products in the high-protein cluster to find the most common protein sources.

```
import re
from collections import Counter

high_protein = plot_df[plot_df['proteins_100g'] >= HIGH_PROTEIN_THRESHOLD]

# Simple candidate protein-source keywords to search for – expand as needed
PROTEIN_SOURCES = ['whey', 'soy', 'pea protein', 'peanut', 'almond', 'egg', 'milk protein', 'casein', 'rice protein']

source_counts = Counter()
for text in high_protein['ingredients_text'].dropna().str.lower():
    for source in PROTEIN_SOURCES:
        if source in text:
            source_counts[source] += 1

pd.Series(source_counts).sort_values(ascending=False).head(3)
```

	0
soy	1992
whey	471
peanut	313

dtype: int64

7. Candidate's Choice

Add one additional chart, filter, or metric here that wasn't asked for, and explain why it adds value.

Ideas: sugar-to-fiber health score, price-per-gram-of-protein, country/brand breakdown.

```
# Flag products containing soy anywhere in the full cleaned dataset (not just high-protein cluster)
clean['contains_soy'] = clean['ingredients_text'].str.lower().str.contains('soy', na=False)

# 1. Soy prevalence by category
soy_by_category = clean.groupby('primary_category')['contains_soy'].mean().sort_values(ascending=False)
print("Share of products containing soy, by category:")
print(soy_by_category)

# 2. Does soy actually correlate with higher protein?
print("\nAverage protein (g/100g): soy vs non-soy products")
print(clean.groupby('contains_soy')['proteins_100g'].mean())

# 3. Sugar profile of soy products – are they part of the healthy cluster or not?
print("\nAverage sugar (g/100g): soy vs non-soy products")
print(clean.groupby('contains_soy')['sugars_100g'].mean())
```

Share of products containing soy, by category:

```
primary_category
Bars                0.503704
Cookies & Biscuits  0.476450
Cereal & Granola    0.435320
Candy & Confectionery 0.187997
Chips & Savory Snacks 0.160203
Other               0.077677
Name: contains_soy, dtype: float64
```

Average protein (g/100g): soy vs non-soy products

```
contains_soy
False    8.332404
True     7.948271
Name: proteins_100g, dtype: float64
```

Average sugar (g/100g): soy vs non-soy products

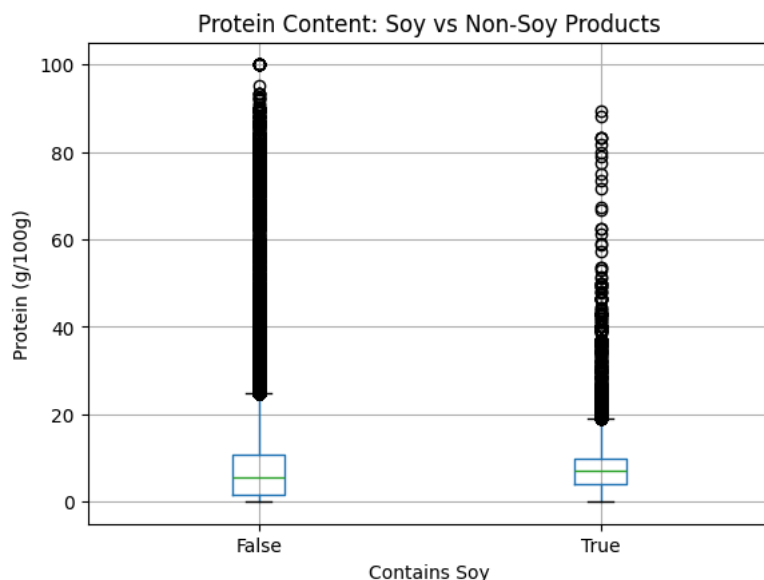
```
contains_soy
False    12.615945
True     14.388967
Name: sugars_100g, dtype: float64
```

The boxplot confirms the printed averages: soy containing products (True) show no meaningful protein advantage over non soy products (False), reinforcing that soy's prevalence in the ingredient list reflects common additive use (lecithin, oil) rather than deliberate protein fortification. This nuance matters for R&D, simply listing "soy" on a label is not evidence of a high-protein product.

```
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))
clean.boxplot(column='proteins_100g', by='contains_soy')
plt.title('Protein Content: Soy vs Non-Soy Products')
plt.suptitle('')
plt.xlabel('Contains Soy')
plt.ylabel('Protein (g/100g)')
plt.show()
```

<Figure size 800x500 with 0 Axes>



While soy appeared most frequently among ingredient lists in the high-protein cluster, a direct comparison shows soy containing products do not actually have higher average protein than non soy products (7.95g vs 8.33g), and trend slightly higher in sugar. This suggests soy's prevalence reflects its ubiquity as a general additive (lecithin, oil) rather than deliberate protein fortification. This is a caution for R&D: simply adding 'soy' to an ingredient list is not a differentiator, the market opportunity lies in products that combine genuinely high protein content with low sugar, regardless of which specific protein source is used.

8. Export Cleaned Data for Dashboard

```
dashboard_cols = ['product_name', 'primary_category', 'sugars_100g', 'proteins_100g']
dashboard_df = clean[dashboard_cols]

dashboard_df.to_csv('cleaned_snacks.csv', index=False)
from google.colab import files
files.download('cleaned_snacks.csv')

import os
print(f"File size: {os.path.getsize('cleaned_snacks.csv') / 1e6:.2f} MB")

File size: 5.49 MB
```

9. Refined Quadrant Analysis (Excluding "Other")

Re-running the quadrant share calculation to confirm results after properly excluding "Other".

```
plot_df = clean[clean['primary_category'] != 'Other'] # make sure Other is excluded here too

HIGH_PROTEIN_THRESHOLD = 10
LOW_SUGAR_THRESHOLD = 5

quadrant_share = plot_df.assign(
    in_empty_quadrant=(plot_df['proteins_100g'] >= HIGH_PROTEIN_THRESHOLD) &
    (plot_df['sugars_100g'] <= LOW_SUGAR_THRESHOLD)
).groupby('primary_category')['in_empty_quadrant'].agg(['mean', 'count'])

quadrant_share.sort_values('mean')
```

	mean	count
primary_category		
Candy & Confectionery	0.012756	4782
Bars	0.055556	1080
Chips & Savory Snacks	0.064209	1573
Cookies & Biscuits	0.086511	3121
Cereal & Granola	0.195070	11642

Final Recommendation

Based on the data, the biggest market opportunity is in Bars, specifically targeting products with at least 10g of protein and less than 5g of sugar per 100g. Despite consumer expectations that bars are a healthy, protein forward category, only 5.6% of Bar products in the market currently meet this threshold — signaling a genuine, under-served gap Helix's client could credibly fill.